Khostty — State & Forward Docket v1.0

Product / subject / beneficiary: `KooshaPari/Khostty` — a Phenotype fork of `ghostty-org/ghostty`. Beneficiary scope: a developer embedding a VT parser, and an agent driving a running terminal over IPC. **Snapshot ID:** `kh-2026-10-17T15:27-0700-8e86ec93a` · **Previous / baseline:** `kh-2026-10-17T05:18Z-4c9e4c57a` **Observed:** 2026-10-17T14:50Z–15:27Z PDT · **Generated:** 2026-10-17T15:27-07:00 **Source:** `main` @ `8e86ec93a1fd0f0c18f39b5c2d9ad604c1ffbaca` · **Worktree:** clean on `main` **Installed / deployed:** `v0.1.0` artifacts exist (crates.io, npm, macOS `.app`, GTK `.deb`, Windows `setup.exe`); **not re-verified this snapshot** → see limitations. **Contract:** State & Forward Docket v1.0 (`handbook/specs/assessment/state-forward-docket-v1/CONTRACT.md`, branch `docs/high-velocity-state-forward-docket-20261016` @ `05e163820`). **Typed record:** `snapshot.json` (canonical). This Markdown is the authored reading projection.

> **Publication status.** Local artifact: yes. Pushed branch: yes (`docs/state-forward-docket-v1-20261017`, PR #51). Merged default: **no**. Deployed: **no**. Safari: **OPENED** (`open -a Safari` on the rendered PDF, this run). A pushed branch is not a merge, and a merge is not an accepted product.

0. Cockpit and publication identity

Field	Value
Product / subject	`khostty` @ `khostty-main-8e86ec93a1fd0f0c18f39b5c2d9ad604c1ffbaca`
Snapshot / previous / baseline	`kh-2026-10-17T15:27-0700-8e86ec93a` / `kh-2026-10-17T05:18Z-4c9e4c57a` / same
Observed / generated	2026-10-17T14:50–15:27 PDT / 2026-10-17T15:27-07:00
Source / worktree	`main` clean; 11 linked worktrees under `~/.jcode/scratch/`
Current stage / confidence	**Library-first CVP shipped; MVP blocked.** Confidence high on engineering (executed), low on usability (mostly UNKNOWN).
Publication	local ✅ · branch ✅ · PR ✅ (#51) · merged ❌ · deployed ❌ · Safari ✅

Two-minute decision brief

What this is: a Ghostty fork whose reason to exist is an **agent-IPC surface** (12 commands, 7 events) plus a reusable VT library. The intended durable outcome is "a usable terminal or embedded terminal that maintains upstream semantics."

Usable now: `cargo add khostty-vt` and `npm install @khostty/libghostty-vt-wasm` give an instant parser; a signed macOS `Khostty.app` runs and takes keystrokes. **Not usable now:** the agent-IPC socket is **never created** — `Server.bind` is called from no runtime (executed check, `E.ipc.bind`). That is the first real blocker and it affects every agent/embedding consumer.

Change since the prior snapshot: prettier drift cleared on 8 PR branches; PR #50 opened for `main`'s drift; session evidence committed (`b6977ebca`); this docket published. **Regression:** none observed. **New uncertainty:** none — the previous snapshot's gaps were confirmed, not reduced.

Next best action: merge the 8 cleared branches + PR #41/#44/#50 (operator, ~1h wall-clock). **Operator decision required:** Q1 (Phase-1 runtime), because it gates the only path to MVP.

Front-page panel

```
ENGINEERING      P=3 F=5 U=5  pass 0.375  coverage 0.615  verified 0.231
USABLE JOURNEYS  P=0 F=4 U=4  pass 0.000  coverage 0.500  verified 0.000
ASSURANCE        P=1 F=2 U=3  pass 0.333  coverage 0.500  verified 0.167
-------------------------------------------------------------------
PRODUCT (unique) P=4 F=11 U=12  pass 0.267  coverage 0.556  verified 0.148
CRITICAL GATE:   BLOCKED  (eng.ipc.runtime.bound, use.journey.agent.ping)
REAL USER VALIDATION: FAIL (zero non-operator users)
```

Denominators are **not** zero. Definitions: assessed pass rate = P/(P+F); coverage = (P+F)/(P+F+U); verified satisfaction = P/(P+F+U). Unknown weight stays in the denominator.

Reading routes

Owner: cockpit → usability → forward path → decisions. Engineer: intent → architecture → gaps → tests → handoff. Reviewer: baseline → evidence → contradictions → critical gates → receipts.

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1. Delta Tree and movement ledger

Baseline `kh-2026-10-17T05:18Z-4c9e4c57a` → current. Membership unchanged (no scope restatement), so a like-for-like delta is admissible.

```
ENGINEERING   [######----] 0.375   retained 3/13   earned 0   lost 0   remaining 10/13
USABLE        [----------] 0.000   retained 0/8    earned 0   lost 0   remaining 8/8
ASSURANCE     [###-------] 0.333   retained 1/6    earned 0   lost 0   remaining 5/6
-------------------------------------------------------------------------------------
PRODUCT       [###-------] 0.267   retained 4/27   earned 0   lost 0   remaining 23/27
```

Movement	Detail
Earned	No new *scored* leaf passed this snapshot. The prettier fix is real but is not a scored leaf; it changes an operator-side merge cost, not a product capability.
Lost / stale	None.
New uncertainty	None.
Reverted / parked	`pane.write` correct repro — parked pending #48 resolution.
Unmeasured	Zig suite, conformance, release assets, GUI journeys — all UNKNOWN, carried in the denominator.

The prettier sweep and the 8 branch pushes are recorded as **work done** in §13/§16, not as score.

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2. Intent, mature contract, CVP and MVP

Mature contract (cited): PhenoRegistry dossier `products/Khostty/DOSSIER.md` — "*a usable terminal or embedded terminal that maintains the required semantics of its upstream.*"

Stage	Definition for this product	Status
CVP (Closest Viable Product)	The smallest living slice preserving identity and surviving an end-to-end journey: a consumable VT parser (Rust crate / WASM / C ABI) plus a runnable desktop app.	**SHIPPED** — `v0.1.0` tag `7e1684f60`
MVP	CVP + the agent-IPC runtime bound on ≥1 platform, so an agent can complete `ping→pong` and `pane.list`.	**NOT DELIVERED**
Mature	MVP + Windows GUI + Phase-1 mux shell.	**BLOCKED** on Q1 + hosts

Authorized exclusions (not passes): Windows GUI runtime (G3 program deferral), `bench/` (G8), upstream-core focus-by-id / synchronous-scan APIs (FORK-ONLY binding), pytest 3.8/3.9 CVE lane, Phase D mux implementation (blocked on Q1). Exclusions are recorded, not scored.

3. Source coverage, archaeology and authority

Input	Authority class	Pinned
Human intent / exclusions	direct user requirement + accepted amendment	operator 2026-10-15 fork-only binding
Repository boundary	PhenoRegistry boundary/SSOT	`KooshaPari/Khostty` only
Accepted design	audit-archive + mux-design session docs	`869b9c75e`, `5708d2ba2`, `72ba2148f`
What the code does	this snapshot's executed checks	`E.*` evidence in `snapshot.json`
Prior assessment	`docs/audits-archive/2026-10-17-final/`	`869b9c75e` (inherited, not re-verified)

Inspected this snapshot: `main` tree; the audit-archive; the two mux-design session docs; all 11 open PRs' metadata; all 10 open issues; `AGENTS.md`; `docs/changelog/0.1.0.md` head; `src/apprt/windows/*` presence; prettier state over 211 tracked files. **NOT inspected:** Zig/conformance runs, GTK source, macOS app source, release asset bytes, PyPI/Go registry state, upstream subtrees. Claims about them are **SOURCE_REPORT**, not executed proof.

4. Current usability and beneficiary journeys

Journey	Beneficiary	Artefact / environment	Result	Evidence
J1 parser via `cargo add khostty-vt`	Rust developer	crates.io 0.1.0 / local	**NOT RUN this snapshot**	—
J2 parser via npm WASM	web developer	npm 0.1.0	**NOT RUN**	—

Journey	Beneficiary	Artefact / environment	Result	Evidence
J3 macOS app launch + keystroke	operator	signed `.app`	**NOT RUN** (prior run 2026-09-20)	—
J4 Linux GTK GUI launch	operator	`.deb`	**NOT RUN** (prior: Xvfb only)	—
J5 Windows GUI window	operator	`setup.exe`	**FAIL** — never opened	`E.win.files`
J6 `ping`→`pong` against live instance	agent	`khostty.sock`	**FAIL** — socket never created	`E.ipc.bind`
J7 conformance corpus	developer	`conformance/build.sh`	**NOT RUN**	`S.conf.8484`

Implementation vs usability vs validation, kept separate:

- **Implementation:** substantial — 7,230 LOC of IPC, a full binding set, CI, docs.
- **Usability:** one class of journey is *believed* usable (parser + macOS app) but **not re-proven this snapshot**; the agent journey is provably blocked.
- **Validation: zero.** No person other than the operator has used any artifact. Scored FAIL, not UNKNOWN.

5. Implementation, architecture and deployment reality

```
graph TB
  VT["VT engine (inherited)"] --> SURF["Surface.zig"]
  SURF --> EMB["embedded / AppKit"]
  SURF --> GTK["gtk"]
  SURF --> WIN["windows (scaffold)"]
  VT --> C["C ABI libghostty-vt"] --> RS["Rust"] & GO["Go"] & PY["Python"]
  VT --> W["WASM"]
  SRV["ipc/server.zig (1,238 LOC)"] -. "NOT BOUND" .-> HOST["ipc/app_host.zig"]
  HOST --> SURF
```

Text equivalent: the terminal engine feeds a render-ready surface, consumed by the app runtimes and the language bindings. The IPC server is compiled and tested but has no edge into any runtime; the dotted edge is the fork's single unfinished link.

Deployment reality: artifacts are **built and published** (`v0.1.0`); installation is verified for macOS/Linux/Windows; **only macOS has an observed GUI launch with input**. Notarization is absent.

6. Progress, assessment coverage and critical gates

Axis	P	F	U	pass rate	coverage	verified
engineering	3	5	5	0.375	0.615	0.231
usable	0	4	4	0.000	0.500	0.000
assurance	1	2	3	0.333	0.500	0.167
product (unique leaves)	**4**	**11**	**12**	**0.267**	**0.556**	**0.148**

Critical gates (cannot be averaged away):

Gate	Status	Why critical
`eng.ipc.runtime.bound`	**FAIL**	without it the fork's differentiating capability is unreachable
`use.journey.agent.ping`	**FAIL**	the user-visible consequence of the above
`ass.real.user.validation`	**FAIL**	nothing is accepted by a real consumer

gate_status: BLOCKED.

7. Stubs, mocks, immature work and undeclared gaps

ID	Location	Present behaviour	Consequence	Closure oracle
S-1	`ipc/app_host.zig` — `Server.bind` unreferenced	socket never created	agent IPC unreachable	booted runtime creates socket; `ping→{"ok":true}`
S-2	`ipc/app_host.zig:290–298`	`pane.write` touches parser cross-thread	race under load; #46 repro premise wrong (#48)	`queueIo` hop lands; correct repro passes
S-3	`apprt/windows/*`	all windowing + IPC `error.Unimplemented`	no Windows window	GUI launches; S1-S6, S8 close
S-4	`apprt/windows/renderer.zig` (64 ln)	orphaned, 8 `Unimplemented`	dead code (A3 pending in #41)	file deleted
S-5	`apprt/windows/win32api.zig` (85 ln)	declares pointers, nothing invokes	dead code (A7 pending in #41)	file deleted
S-6	`app_host.zig:177–178,208–232`	`pane.focus/pane.search` → `error.Unsupported`	no focus-by-id / sync scan	disclosed in docs (A1)
S-7	`pkg/opengl/egl.zig:470–473`	tautological assertion — false control	a broken EGL arm passes CI	A2 real assertion (#41)
S-8	`terminal/search/Thread.zig:197`	`.BADF` → `unreachable` panic	suite ABRTs on a race	A5 teardown guard (#41)
S-9	`wasm/package.json`	version `0.0.0`	drift vs npm 0.1.0	A6 bump (#41)
S-10	5 doc claims	understate the code	plans over-estimate work	A1 refresh (#41)
S-11	`fake_host.zig` (446 ln)	test-only backend	none — by design	n/a

8. Assurance, privacy, integrity and recovery

Aspect	State
IPC auth	fail-closed token, constant-time SHA-256 (source-read; **not executed this snapshot**)
Rate limiting	absent (B3 adds it; in #44)
Privacy / data egress	none observed — no telemetry, no remote sink in the inspected tree
Integrity	macOS `.app` signed, **not notarized**; hashes verified at release time
Recovery	`.deb` `dpkg -r`; Windows uninstaller verified; `cargo update --precise` for the crate
Negative controls	the EGL assertion was a **false** control (S-7)
Independent re-run	**not performed** — a required gate for acceptance

9. UX, accessibility, developer experience and documentation

- **UX:** macOS app behaves like upstream (prior run); Windows no window; GTK partial.
- **AX:** inherited, **never audited in the fork** — an open obligation, not deferred.
- **DX:** `cargo add` / `npm install` are the documented entry points; Go has `examples/`.
- **Docs:** 14 hand-curated docs; 5 stale claims being refreshed by A1 (#41).

10. Research coverage, alternatives and decisions

Comparator	Status
Upstream Ghostty (pinned)	tracked; last sync #18 (397 commits)
WezTerm / Kitty / native terminal	shortlist seeded, **not evaluated**
`alacritty` / `foot` / `tmux`	adjacent, **not evaluated**

Advantage hypothesis: "a specific embedding/workflow improvement with no unacceptable terminal correctness regression." The agent-IPC surface is that improvement; it is **not yet evaluated** because it is not yet runnable. Honest status, not a claim.

11. Fork, rewrite, migration and integration completion

Aspect	State
Type	**Fork** (not rewrite)

Aspect	State
Upstream	`ghostty-org/ghostty`; merge base `b3dc3d794`
Divergence	**307 ahead / 3 behind** (upstream `e3ebb1d1b`)
Delta	352 files, +74,959 / −233
Retained	terminal engine, renderers, macOS/GTK runtimes, CLI IPC
Changed	agent-IPC subsystem, fork docs, CI, Windows scaffold
License	upstream MIT inherited; no relicensing claimed
Migration	n/a — no fork-owned persisted data format
Protected paths	`src/terminal/`, `src/Surface.zig`, `src/Command.zig`, `src/runner.zig` (require `// FORK:` + issue)

12. Forward path, critical path and time to useful completion

Work ID	Outcome / closure oracle	Depends on	Owner	Agent effort	Wall-clock	Confidence
W1	8 prettier-cleared branches merge green	operator merge	operator	0	0.5–2h	high
W2	Phase A (A1-A7) on `main` (#41)	operator merge	operator	0	1–3h	high
W3	B3/B6 + C1-C4 on `main` (#44)	operator merge	operator	0	1–3h	high
W4	#48 resolved; correct `pane.write` repro	—	agent	3–6h	+1d	medium
W5	**B1**: bind `Server` from GTK	GTK host	agent	4–8h	+1–2d	medium
W6	**B2**: `pane.write` `queueIo` hop	W5	agent	3–6h	+1d	medium
W7	B4/B5: event hooks + counters	W5	agent	4–8h	+1–2d	medium
W8	S1-S6, S8: Windows GUI	Windows rig	agent+operator	days	+1–2w	low
W9	Accessibility audit	—	agent	2–4h	+1d	medium

Required horizons

- **Next 30 min:** land W1 (merge the 8 cleared branches) — highest-value unblock, no new code.
- **Next 1 h:** land W2 + W3 so `main` carries the polish and hardening; fallback = W4 in parallel.
- **Next 4 h:** W4 (correct race repro) authored and CI-run.
- **Next 24 h:** W5 attempted if a GTK host exists; otherwise publish the honest blocker.

- **CVP:** already shipped. **MVP:** 1–4 days **after** a GTK host exists (agent effort ~1–2d; the unknown is host availability, not code). **Mature:** 2–8 weeks, dominated by Q1, the Windows rig, and Phase D.

Forecast basis: calibrated to this fork's own cycle times (PR #15 Phase-0 build wiring ~6d branch→merge; the A1-A7 batch authored in ~3h in one run). No P50/P90 is claimed. **Critical path is gated by operator-provided hosts and one decision, not by agent throughput.** Replan when Q1 or host availability changes.

13. Decisions, blockers and operator actions

Open decisions: Q1 Phase-1 runtime (recommend **option (c) mock-only** per `06_Q1_DECISION_MATRIX.md`); Q2 GTK commitment; Q3 dependency policy; Q4 Windows timing.

Operator actions: merge W1/W2/W3; decide #48 vs #46; provide/reach a GTK host; free ≥50 GiB on the build host; set `SONAR_TOKEN`; configure PyPI trusted publishing; notarize the macOS app.

Closed this arc: D1 Phase-0 merged (`ebc033eba`); D2 fork-only binding (2026-10-15); D3 Phase A authored (#41); D4 prettier drift cleared on 8 branches + PR #50 (this run).

14. Operations, lifecycle, cost and value

Cost centre	Observation
CI wall time	Trunk Check ~9-10 min; Test job long; several workflows cancel-loop
CI noise	`Vouch — Manage by Is…` re-runs in a cancel loop on `main`
Build host	earlier 96-98% disk; release build failed on disk, debug build 16m33s
Maintenance	3 commits behind upstream; sync via PR #18
Value delivered so far	a usable VT parser (crate/npm/C ABI) + a runnable macOS app

15. Risks, contradictions and confidence

Rank	Risk	Severity	Mitigation
R1	Agent-IPC never wired	High	W5 on a GTK host, or ship option (c) honestly
R2	Q1 unmade	High	operator picks (c)
R3	`pane.write` race ships	High	W6 with W5; #48 corrects the repro
R4	Windows GUI never launched	Medium	#30 + S8
R5	Build-host disk	Medium	free ≥50 GiB
R6	AX never audited	Medium	W9

Rank	Risk	Severity	Mitigation
R7	Format drift recurs	Low	PR #50 + scheduled pass

Contradictions found: the README status (Windows "DONE") vs `docs/PLATFORMS.md` ("PARTIAL") — S8 tracks it; the A1 refresh reconciles it. **Confidence:** high on executed engineering; medium on inherited test counts; UNKNOWN where marked.

16. Restart packet, repository sync and publication receipts

Restart packet: repo `/Users/kooshapari/CodeProjects/Phenotype/repos/khostty`; branch `main` @ `8e86ec93a`; contract at `handbook/specs/assessment/state-forward-docket-v1/CONTRACT.md`. First actions: merge W1/W2/W3; read `snapshot.json` for the leaf set; resume W4.

Receipts (this run):

Action	Receipt
Session evidence commit	`b6977ebca` (pushed branch `session-20261017-pr-triage-stop`)
Prettier sweep PR	**#50**
Per-branch prettier commits	#36 `c53694b08` · #39 `cd642a279` · #40 `a58b2335f` · #41 `f1466ca47` · #43 `542962330` · #44 `71c5cd57b` · #46 `b88f0c600+07695a4ec` · #47 `650fef21b`
Docket PR	**#51** (`docs/state-forward-docket-v1-20261017`, `f2a16a011`)
PDF	rendered from the same Markdown source; opened in Safari this run
PhenoRegistry mirror	`audits/khostty-state-forward-docket-2026-10-17/` (this publication)

17. Acceptance, finish, exclusions and sunset

Completion is NOT declared. No named scope has accepted gates + current proof + usable delivery + accurate docs + no hidden in-scope stubs. The critical gate (`eng.ipc.runtime.bound`) fails.

Authorized exclusions (recorded, not passes): Windows GUI runtime; `bench/`; upstream-core APIs; pytest 3.8/3.9; Phase D (pending Q1); scaling/sunset/user-learning.

Sunset: none. The project is active.

18. Evidence index and source-to-claim traceability

Evidence ID	Kind	Locator (sha256 in `snapshot.json`)
`E.main.rev`	EXECUTED_PRODUCT	`git rev-parse main`

Evidence ID	Kind	Locator (sha256 in `snapshot.json`)
`E.main.head`	EXECUTED_PRODUCT	`git log -1 --format='%H %ci %s' main`
`E.fork.delta`	EXECUTED_PRODUCT	`git rev-list --left-right --count origin/main...upstream/main`
`E.fork.shortstat`	EXECUTED_PRODUCT	`git diff --shortstat <merge-base> origin/main`
`E.ipc.loc`	EXECUTED_PRODUCT	`wc -l src/apprt/ipc/*.zig`
`E.ipc.bind`	EXECUTED_PRODUCT	`grep -rn 'Server.bind' src/ --include=*.zig` (minus `ipc/`)
`E.win.files`	EXECUTED_PRODUCT	`ls src/apprt/windows/`
`E.prettier.main`	EXECUTED_PRODUCT	`prettier --check` over 211 tracked files
`E.prs.open` / `E.issues.open`	EXECUTED_PRODUCT	`gh pr/issue list --state open`
`S.audit.archive` / `S.conf.8484`	SOURCE_REPORT	inherited, not re-verified

Full hashes, subjects, observation times and freshness bounds are in `snapshot.json` (canonical).

End of Khostty State & Forward Docket v1.0 — snapshot `kh-2026-10-17T15:27-0700-8e86ec93a`.